The Particle Model — Paper D

HARD · NON-CALCULATOR · 45 MINUTES · 30 MARKS

NAME　　DATE　　SCORE / 30

- Answer every question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says explain, your answer must talk about particles — their arrangement, their movement, the spaces between them, or the forces between them.
- The mark for each question is shown in square brackets on the right.
- Several questions here describe experiments you may not have seen before. You are not expected to recognise them — apply the particle ideas you already have.
- Never write that particles melt, expand, or get bigger. Only their arrangement, spacing and energy change.

SECTION 1 — READING A COOLING CURVE

1. A liquid is allowed to cool and its temperature is recorded every minute.

Time (min):	0	2	4	6	8	10	12	14
Temp (°C):	90	72	55	55	55	55	41	30

(a) State the freezing point of the substance. [1]

(b) State the state of the substance at 2 minutes and at 14 minutes. [2]

(c) Explain why the temperature stays constant between 4 and 10 minutes, even though the substance is still losing energy to the room. [3]

SECTION 2 — THE DIFFUSION TUBE

2. A long glass tube has cotton wool soaked in ammonia pushed into one end and cotton wool soaked in hydrogen chloride pushed into the other, at the same moment. Both give off gases. After a few minutes a white ring of solid forms inside the tube, closer to the hydrogen chloride end.

(a) Explain how the two gases travelled along the tube. [2]

(b) Explain why the ring does not form in the middle. [2]

(c) Predict, with a reason, what would happen to the position of the ring if the experiment were repeated in a warmer room. [1]

SECTION 3 — THE LONG EXPLANATION

3. A block of ice at −20 °C is heated steadily until it has all become steam at 120 °C. Describe and explain the whole journey in terms of particles and energy. Your answer should refer to both flat sections of the heating curve, and to what the energy is doing at each stage. [6]

4. A substance melts at −7 °C and boils at 58 °C.

(a) State its state at 20 °C, and explain how you know. [2]

(b) State its state at 80 °C, and explain how you know. [2]

5. Water is boiled in an open pan on a stove. After ten minutes the pan and its contents have a smaller mass than before.

(a) Explain where the missing mass has gone. [2]

(b) Explain why this does not break the rule that mass is conserved. [2]

6. A student has written two statements in her book, and both are wrong. [5]
(i) “When a solid is heated, the particles get bigger, so the solid expands.”
(ii) “The flat part of the heating curve shows that nothing is happening.”
For each one, explain the mistake and write a correct version.

End of paper. Check every answer has a unit or a form the question actually asked for.